# Course notes

## Add or multiply

For finite pairwise disjoint sets $A_i$, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, we must correct double counting: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of elements of the product Cartesian $A \times B$ is $|A||B|$. For a sequence of choices, multiply the numbers of possibilities at each stage, provided these numbers are valid for each previous choice considered.

**Example.** A code of two letters chosen from 26, followed by three digits, with repetition allowed, can take $26^2 \times 10^3$ values. A three-digit number does not start with 0: there are $9 \times 10^2$.

## Lists and subsets

In a set of $n$ elements, there are $n^k$ ordered lists of length $k$ with repetition; $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ lists of $k$ distinct elements, if $0 \leq k \leq n$ ; $n!$ permutations of all elements. We set $0! = 1$.

**Combinations**

A combination of $k$ elements among $n$ is a subset of $k$ elements; order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

**Why divide by $k!$ ?** Each subset of size $k$ gives exactly $k!$ distinct lists depending on the order of its elements. The number of lists without repetition is therefore $k!\binom{n}{k}$.

## Identities and proofs

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's rule above is used for $0 \leq k < n$ ; the endpoint formulas can be read directly with the values 1.

**Symmetry.** Taking the complement sets up a one-to-one correspondence between the subsets of size $k$ and those of size $n-k$.

**Pascal, combinatorial proof.** In a set of $n+1$ elements, fix one element $a$. A subset of size $k+1$ contains $a$ (choose $k$ other elements from $n$) or does not contain it (choose $k+1$ elements among $n$). The two cases are disjoint and cover every possibility.

**Pascal, algebraic proof.** For $0 \leq k < n$, put over a common denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

Edge cases are dealt with directly.

**Number of subsets.** For each of the $n$ elements, choosing "present" or "absent" gives $2^n$ subsets. Classifying them by size, we count $\sum_{k=0}^{n}\binom{n}{k}$ : both counts are equal.

**Extension mentioned in the scheme of work.** Expanding $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ is understood by choosing, in each of the $n$ factors, the term $a$ or $b$. This formula is not used as a prerequisite for the written tests in this booklet.

## Define what is being counted

### 9.1 Order and repetition: two separate questions

First use: S22.

**Worked example · Two letters chosen from A, B, C**

With order and repetition, the nine lists are AA, AB, AC, BA, BB, BC, CA, CB, CC. With order without repetition, six lists remain. Without order or repetition, the three objects are $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$. The result changes because the counted object changes, not because we choose another formula.

| 1st letter / 2nd letter | A | B | C |
|---|---|---|---|
| A | AA | AB | AC |
| B | BA | BB | BC |
| C | CA | CB | CC |

Cross out the diagonal to forbid repetition; group AB and BA to ignore order.

**A1 · Immediate application** List the codes of two distinct digits taken from $\{0, 1, 2\}$. How many represent two-digit numbers? Explain the difference between a code and a number.

**Property · Addition and product**

Disjoint cases are counted by addition. Successive choices are counted by multiplication if each stage offers the stated number of possibilities, given the previous choices. For $n$ distinct objects, the lists of length $k$ without repetition number $n(n-1)\cdots(n-k+1)$ ; the number of complete permutations is $n!$.

**A2 · Partial guidance** An outfit has one top from four and one pair of trousers from three. One top and a pair of trousers are red. Count the outfits containing both red items, then exactly one red item.

### 9.2 Forgetting order without losing possibilities

First use: S30.

**Definition · Binomial coefficient**

$\binom{n}{k}$ is the number of subsets with $k$ elements of a set with $n$ elements. For $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. In particular $\binom{n}{0} = \binom{n}{n} = 1$.

**Proof.** Each subset with $k$ elements gives $k!$ distinct ordered lists. All the subsets give the same number of lists. Dividing the $n!/(n-k)!$ lists without repetition by $k!$ thus counts each subset exactly once. □

**Worked example · A team and roles**

A team of three students out of eight can be chosen in $\binom{8}{3} = 56$ ways. If one of the three becomes captain, there are $56 \times 3 = 168$ choices. We recover 8 $\binom{7}{2} = 168$ by first choosing the captain. To assign three distinct roles, there would be 8 $\times 7 \times 6 = 336$ possibilities.

**A3 · Independent choice** Ten pupils, six in Terminale and four in Première, form a team of three. Count the teams containing exactly two Terminale students, then at least one student from each level, by two methods.

**A4 · Fix a count** "Choosing two delegates out of five gives 5 $\times$ 4 $=$ 20 possibilities." Explain the mistake if the roles are identical. In which situation would the result 20 be correct?

### 9.3 Two proofs of Pascal's rule, two readings of a sum

First use: S35 and S37.

**Property · Proven identities**

For $1 \leq k \leq n$, $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$. In addition, $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

**Proof.** Single out one element in a set of size $n+1$. The subsets of size $k$ that do not contain it are $\binom{n}{k}$ ; those which contain it require $k-1$ other elements, i.e. $\binom{n}{k-1}$. The two cases are disjoint and exhaustive. By calculation, put the two fractions over the same denominator $k!(n-k+1)!$ gives a numerator $n![(n-k+1)+k] = (n+1)!$. Finally, a subset of a set with $n$ elements is built by $n$ binary choices (take each element or not), i.e. $2^n$ subsets; the sorting them by size gives the sum. □

**Worked example · A distinguished element**

Among five pupils including Lina, a team of two contains Lina (four choices for her partner) or does not contain her ($\binom{4}{2} = 6$ choices). Hence $\binom{5}{2} = 4 + 6 = 10$.

**A5 · Immediate application** From row 1, 4, 6, 4, 1 of Pascal's triangle, construct the next one and check the sum of its coefficients. Interpret the coefficient 10.

### 9.4 Complement and double counting

First use: S35.

**Worked example · At least one occurrence**

A four-digit code in $\{0,1,2,3,4\}$ allows repetition. There are $5^4$ codes and $4^4$ codes without zero, therefore $5^4 - 4^4 = 369$ codes with at least one zero. The calculation $4 \times 5^3$ would count codes containing several zeros several times.

**A6 · Choice of strategy** Among these codes, count those that contain exactly two zeros. Then count those whose digit sum is odd, without listing the 625 codes.

A first step: For the second question, classify according to the number of positions holding an odd digit: one or three.

**One minute, spoken aloud.** Explain why systematically dividing a product by a factorial may be wrong: first state what is regarded as the same result.

**Delayed review · S41 then S85.** Recover the binomial coefficient as a number of paths containing exactly $k$ successes; compare this interpretation with that of a team.

## Investigation

**Order and repetition in counting - 30 min.**

Six different students volunteer to organise a debate.

1. Two pupils are chosen to form a pair without assigning roles. How many pairs can be formed?
2. A president and a secretary are appointed, who must be two different pupils. How many of assignments of the two roles are possible?
3. Three successive presentations are planned. The same pupil may present several times. How many schedules of three presentations are possible?
4. We start again with only four pupils, denoted by $A$, $B$, $C$ and $D$. Write all pairs and all the assignments of the two roles. Then explain why there are 64 schedules of three presentations, without listing them.
5. For each of the three ways of choosing, state whether order matters and whether the same person can be chosen several times.